

NAT-ICS Practice Test

National Aptitude Test — Computer Science Group

90 MCQs • 120 Minutes

Test Pattern (Official NTS NAT-ICS Distribution)

Section	Sub-section	MCQs
A. English	Sentence Completion (5) + Analogy (3) + Antonym (4) + Comprehension (5) + Synonym (3)	20
B. Analytical	Scenario-Based (12) + Statements-Based (8)	20
C. Quantitative	Arithmetic (13) + Algebra (4) + Geometry (3)	20
D. Subject	Physics (10) + Computer (10) + Mathematics (10)	30
Total		90

Scoring: +1 correct, −0.25 wrong, 0 skipped. **Document structure:** Part I – all questions.
Part II – worked answers with reasoning.

PART I

Questions (Q1–Q90)

1. English

A1. Sentence Completion (Q1–Q5)

Q1.

Despite the heavy rain, the team _____ to complete the project on time.

- (A) managed (B) manages
(C) managing (D) manage

Q2.

She is _____ honest person that everyone trusts her with confidential information.

- (A) so (B) such
(C) such an (D) such a

Q3.

The committee will announce the results _____ the meeting concludes.

- (A) as soon as (B) no sooner
(C) hardly when (D) while

Q4.

If he _____ harder, he would have passed the entrance test easily.

- (A) studies (B) studied
(C) had studied (D) has studied

Q5.

The new policy is intended to _____ poverty in rural areas.

- (A) alleviate (B) elevate
(C) aggravate (D) advocate

A2. Analogy (Q6–Q8)

Q6.

DOCTOR : HOSPITAL ::

- (A) Teacher : Student (B) Chef : Restaurant
(C) Engineer : Bridge (D) Lawyer : Client

Q7.

PEN : WRITE ::

- (A) Book : Read (B) Knife : Cut
(C) Bed : Sleep (D) Car : Drive

Q8.

BIRD : FLOCK ::

- (A) Tree : Forest (B) Wolf : Pack
(C) Star : Galaxy (D) Ant : Colony

A3. Antonyms (Q9–Q12)**Q9.**

BENEVOLENT

- (A) Kind (B) Malevolent
(C) Generous (D) Charitable

Q10.

CONCISE

- (A) Brief (B) Verbose
(C) Clear (D) Direct

Q11.

TRANSPARENT

- (A) Clear (B) Visible
(C) Opaque (D) Translucent

Q12.

ASCEND

- (A) Climb (B) Rise
(C) Descend (D) Soar

A4. Reading Comprehension (Q13–Q17)

Passage: The advent of artificial intelligence has fundamentally altered how organizations approach data-driven decision-making. Machine learning algorithms, once confined to academic research, are now embedded in everyday applications — from recommendation systems on streaming platforms to fraud detection in banking. However, this rapid adoption has raised concerns about algorithmic bias, data privacy, and the displacement of human labour. Critics argue that without transparent oversight, AI systems may perpetuate the very inequalities they were designed to remove. Proponents counter that, when carefully audited, these systems can deliver outcomes that are fairer and more consistent than purely human judgement. The challenge for policymakers is to craft frameworks that capture the benefits of AI while mitigating its risks.

Q13.

The main idea of the passage is best expressed by:

- (A) AI is universally beneficial to society. (B) AI presents both opportunities and challenges that require thoughtful regulation.
(C) AI will inevitably replace human workers. (D) AI research should be confined to academia.

Q14.

According to the passage, critics of AI are mainly concerned about:

- (A) The cost of deploying AI systems. (B) The speed of AI computation.
(C) Bias, privacy, and job displacement. (D) The complexity of machine learning algorithms.

Q15.

The word *perpetuate* in the passage most nearly means:

- (A) Eliminate (B) Continue or maintain
(C) Investigate (D) Discover

Q16.

The author's tone in the passage can best be described as:

- (A) Strongly opposed to AI (B) Enthusiastically supportive
(C) Balanced and analytical (D) Sarcastic

Q17.

Which statement would the author most likely agree with?

- (A) AI systems should be banned. (B) AI auditing is unnecessary for fair outcomes.
(C) Policy frameworks should balance benefits and risks of AI. (D) Human judgement is always superior to algorithmic decision-making.

A5. Synonyms (Q18–Q20)**Q18.**

METICULOUS

- (A) Careless (B) Thorough
(C) Lazy (D) Hasty

Q19.

AMBIGUOUS

- (A) Clear (B) Certain
(C) Unclear (D) Loud

Q20.

TENACIOUS

- (A) Weak (B) Persistent
(C) Timid (D) Fragile

2. Analytical Reasoning

B1. Scenario-Based Questions (Q21–Q32)

Scenario 1 (Q21–Q24). Six students — A, B, C, D, E, F — sit in a row facing north. Conditions:

- A sits to the immediate right of D.
- C is at one of the extreme ends.
- B sits between E and F.
- D is third from the left end.
- F is not adjacent to C.

Q21.

Who sits at the extreme left end? (A) E (B) F
(C) B (D) Cannot be determined

Q22.

Which pair sits adjacent to each other? (A) A and C (B) D and B
(C) E and B (D) F and C

Q23.

If F and C exchange places, who is fourth from the right?
(A) D (B) B
(C) F (D) A

Q24.

How many students sit between A and C? (A) One (B) Two
(C) Three (D) None

Scenario 2 (Q25–Q28). Five tasks — Testing (T), Coding (C), Design (D), Review (R), Deployment (P) — on Mon–Fri, one task per day. Constraints:

- Design before Coding.
- Testing immediately follows Coding.
- Deployment is last.
- Review on Tuesday.

Q25.

Which task is scheduled on Monday? (A) Coding (B) Design
(C) Testing (D) Review

Q26.

Coding is performed on: (A) Tuesday (B) Wednesday
(C) Thursday (D) Friday

Q27.

Testing is performed on: (A) Wednesday (B) Thursday
(C) Friday (D) Monday

Q28.

Correct order from Monday to Friday? (A) D, R, C, T, P (B) R, D, C, T, P
(C) D, C, R, T, P (D) D, R, T, C, P

Scenario 3 (Q29–Q32). Five friends — P, Q, R, S, T — each prefer a different language from {Python, Java, C++, JavaScript, Go}. Constraints:

- P does not like Java or C++.
- Q likes Java.
- R likes either Python or Go.
- S likes JavaScript.
- T does not like Python.

Q29.

Which language does T like? (A) Python (B) Go
(C) C++ (D) Java

Q30.

Which language does R like? (A) Python (B) Go
(C) Either Python or Go (D) Java

Q31.

If R likes Go, then P likes: (A) Java (B) C++
(C) Python (D) JavaScript

Q32.

Who likes JavaScript? (A) P (B) Q
(C) R (D) S

B2. Statements-Based Questions (Q33–Q40)

Q33.

Statements: All engineers are graduates. Some graduates are professors. Conclusion: Some engineers are professors.
(A) Conclusion follows (B) Conclusion does not follow
(C) Cannot be determined (D) Both engineers and professors are graduates

Q34.

Statements: All birds can fly. Penguins are birds. Conclusion: Penguins can fly.

- (A) Conclusion follows logically from the premises
 (B) Conclusion does not follow
 (C) Penguins are not birds
 (D) Birds cannot fly

Q35.

- If COMPUTER $\rightarrow$ DPNQVUFS, then PROGRAM $\rightarrow$?
 (A) QSPHSBN (B) QSPHSAN
 (C) QSPGSBN (D) QSPHRAM

Q36.

- A + B means A is father of B; A - B means A is sister of B; A $\times$ B means A is brother of B. In P + Q - R, P is related to R as:
 (A) Father (B) Uncle
 (C) Brother (D) Grandfather

Q37.

- Find the next number in the series: 2, 6, 12, 20, 30, ?
 (A) 40 (B) 42
 (C) 44 (D) 48

Q38.

- Pointing to a man, Sara said, "He is the son of my mother's only son." How is the man related to Sara?
 (A) Brother (B) Cousin
 (C) Nephew (D) Son

Q39.

- If WATER $\rightarrow$ YCVGT, then EARTH $\rightarrow$?
 (A) GCTVJ (B) GCVTJ
 (C) GBTVJ (D) GCTUJ

Q40.

- Statements: No student is lazy. All hardworking people are students.
 Conclusion: No hardworking person is lazy.
 (A) Conclusion follows (B) Conclusion does not follow
 (C) Data is insufficient (D) Conclusion is partially true

3. Quantitative Reasoning

C1. Arithmetic (Q41–Q53)

Q41.

If 30% of a number is 60, then 50% of the same number is:
(A) 80 (B) 90
(C) 100 (D) 120

Q42.

A shopkeeper marks an item at Rs. 800 and offers a 15% discount. SP =?
(A) Rs. 680 (B) Rs. 700
(C) Rs. 720 (D) Rs. 750

Q43.

The average of five consecutive integers is 17. The largest is:
(A) 17 (B) 18
(C) 19 (D) 20

Q44.

A train travels 240 km in 4 hours. Speed in km/h:
(A) 50 (B) 55
(C) 60 (D) 65

Q45.

Boys:Girls = 3 : 2, with 30 boys. Total students =?
(A) 40 (B) 45
(C) 50 (D) 60

Q46.

Simple interest on Rs. 5000 at 8% p.a. for 3 years: (A) Rs. 1000 (B) Rs. 1200
(C) Rs. 1500 (D) Rs. 1800

Q47.

Bought at Rs. 1200, sold at Rs. 1500. Profit %? (A) 20% (B) 25%
(C) 30% (D) 33%

Q48.

HCF of 24 and 36: (A) 6 (B) 8
(C) 12 (D) 18

Q49.

A worker completes a job in 12 days. In 4 days he completes:
(A) $\frac{1}{4}$ (B) $\frac{1}{3}$
(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

Q50.

LCM of 6, 9, 12: (A) 18 (B) 24
(C) 36 (D) 72

Q51.

A sum doubles itself at simple interest in 8 years. Rate =?
(A) 10% (B) 12.5%
(C) 15% (D) 20%

Q52.

8 men do a work in 12 days; 6 men will do the same in:
(A) 14 days (B) 16 days
(C) 18 days (D) 20 days

Q53.

Price increased 20%, then decreased 20%. Net change?
(A) No change (B) 4% decrease
(C) 4% increase (D) 2% decrease

C2. Algebra (Q54–Q57)**Q54.**

If $x+5 = 12$, then $x^2 = ?$ (A) 49 (B) 36
(C) 25 (D) 144

Q55.

Solve $2x-3 = 7$. (A) 3 (B) 4
(C) 5 (D) 6

Q56.

Roots of $x^2-5x+6 = 0$: (A) 1 and 6 (B) 2 and 3
(C) -2 and -3 (D) 1 and 5

Q57.

If $a = 3$ and $b = -2$, then $a^2 + b^2 - 2ab = ?$
(A) 1 (B) 13
(C) 25 (D) -1

C3. Geometry (Q58–Q60)**Q58.**

Sum of interior angles of a triangle: (A) 90° (B) 180°
(C) 270° (D) 360°

Q59.

Area of a rectangle with $\ell = 8$ cm, $w = 5$ cm:

(A) 13 cm^2 (B) 26 cm^2
(C) 40 cm^2 (D) 45 cm^2

Q60.

Circumference of a circle with $r = 7$ cm, taking $\pi = 22/7$:

(A) 22 cm (B) 44 cm
(C) 49 cm (D) 154 cm

4. Subject – Physics, Computer, Mathematics

D1. Physics (Q61–Q70)

Q61.

SI unit of electric current: (A) Volt (B) Ampere
(C) Ohm (D) Watt

Q62.

Acceleration due to gravity on Earth's surface: (A) 8.9 m/s^2 (B) 9.8 m/s^2
(C) 10.8 m/s^2 (D) 11.2 m/s^2

Q63.

Which is a scalar quantity? (A) Velocity (B) Force
(C) Displacement (D) Speed

Q64.

Ohm's law: $V = IR$. If $V = 12 \text{ V}$ and $R = 4 \Omega$, then $I = ?$
(A) 2 A (B) 3 A
(C) 4 A (D) 6 A

Q65.

Work done by 10 N moving an object 5 m in the direction of force:
(A) 20 J (B) 30 J
(C) 50 J (D) 100 J

Q66.

Bending of light when passing between media is called:
(A) Reflection (B) Refraction
(C) Diffraction (D) Dispersion

Q67.

Unit of power: (A) Joule (B) Newton
(C) Watt (D) Pascal

Q68.

Which is a vector quantity? (A) Mass (B) Temperature
(C) Velocity (D) Time

Q69.

Frequency of AC supply in Pakistan: (A) 50 Hz (B) 60 Hz
(C) 100 Hz (D) 220 Hz

Q70.

KE of a 2 kg object moving at 4 m/s: (A) 8 J (B) 16 J
(C) 32 J (D) 64 J

D2. Computer (Q71–Q80)*D2.1 Introduction to Computer***Q71.**

An example of system software: (A) MS Word (B) Operating System
(C) Adobe Photoshop (D) Google Chrome

Q72.

1 Kilobyte (KB) equals: (A) 1000 bytes (B) 1024 bytes
(C) 1024 bits (D) 100 bytes

Q73.

Device that converts digital signals to analog and vice versa:
(A) Router (B) Modem
(C) Hub (D) Switch

Q74.

CPU consists primarily of: (A) ROM and RAM (B) Hard disk and Monitor
(C) Control Unit and ALU (D) Keyboard and Mouse

*D2.2 Programming Skills***Q75.**

Which is NOT a programming language? (A) Python (B) Java
(C) HTML (D) C++

Q76.

A variable in programming is: (A) A fixed value that does not change (B) A named storage location that value which can change
(C) A type of loop (D) A function that returns no value

Q77.

Symbol for a single-line comment in C/C++: (A) # (B) //
(C) /* */ (D) <!-->

*D2.3 Fundamentals of Database***Q78.**

DBMS stands for: (A) Data Backup Management System (B) Database Management System
(C) Digital Binary Memory System (D) Database Modeling Syntax

Q79.

A primary key in a relational database is: (A) A key that can be NULL (B) A key used only for foreign keys
(C) A unique identifier for each row in a table (D) A duplicate identifier

Q80.

SQL command to retrieve data: (A) INSERT (B) UPDATE
(C) DELETE (D) SELECT

D3. Mathematics (Q81–Q90)**Q81.**

$\sin 30^\circ =$ (A) 0 (B) $1/2$
(C) $\sqrt{3}/2$ (D) 1

Q82.

Derivative of x^3 with respect to x : (A) $3x$ (B) x^2
(C) $3x^2$ (D) $x^3/3$

Q83.

If $f(x) = 2x+3$, then $f(4) = ?$ (A) 8 (B) 9
(C) 11 (D) 14

Q84.

$\log_{10} 1000 =$ (A) 1 (B) 2
(C) 3 (D) 10

Q85.

$\int 2x \, dx =$ (A) $x^2 + C$ (B) $2 + C$
(C) $2x^2 + C$ (D) $x + C$

Q86.

Modulus of the complex number $3+4i$: (A) 3 (B) 4
(C) 5 (D) 7

Q87.

Slope of the line through (1, 2) and (3, 8): (A) 2 (B) 3
(C) 4 (D) 6

Q88.

Number of arrangements of the letters in EXAM: (A) 12 (B) 24
(C) 36 (D) 48

Q89.

If A is 2×3 and B is 3×4 , then AB has dimensions:
(A) 2×3 (B) 3×4
(C) 2×4 (D) Not defined

Q90.

$\cos 0^\circ + \sin 90^\circ =$ (A) 0 (B) 1
(C) 2 (D) -1

PART II

Answers & Worked Solutions

Quick Reference Key

Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans	Q#	Ans
1	A	19	C	37	B	55	C	73	B
2	C	20	B	38	D	56	B	74	C
3	A	21	A	39	A	57	C	75	C
4	C	22	C	40	A	58	B	76	B
5	A	23	A	41	C	59	C	77	B
6	B	24	A	42	A	60	B	78	B
7	B	25	B	43	C	61	B	79	C
8	B	26	B	44	C	62	B	80	D
9	B	27	B	45	C	63	D	81	B
10	B	28	A	46	B	64	B	82	C
11	C	29	C	47	B	65	C	83	C
12	C	30	C	48	C	66	B	84	C
13	B	31	C	49	B	67	C	85	A
14	C	32	D	50	C	68	C	86	C
15	B	33	B	51	B	69	A	87	B
16	C	34	A	52	B	70	B	88	B
17	C	35	A	53	B	71	B	89	C
18	B	36	A	54	A	72	B	90	C

1. English – Worked Answers

Q1. Answer: A (managed)

Past tense is required to agree with the completed action implied by “Despite the heavy rain.” Options C and D are grammatically incompatible; B is present tense and disagrees with the temporal frame.

Q2. Answer: C (such an)

The structure is “such + a/an + adjective + noun + that.” “Honest” begins with a vowel sound, so the article is *an*, not *a*.

Q3. Answer: A (as soon as)

“No sooner” requires “than”; “hardly when” is paired idiom; “while” implies simultaneity, not sequence. “As soon as” fits cleanly.

Q4. Answer: C (had studied)

Third conditional: *If + past perfect, would have + past participle*.

Q5. Answer: A (alleviate)

Alleviate = lessen, reduce. Elevate = raise, aggravate = worsen, advocate = support — only A fits “poverty reduction.”

Q6. Answer: B (Chef : Restaurant)

Professional : workplace. A doctor works in a hospital like a chef works in a restaurant. The others mix in object-of-work or person-with-whom relations.

Q7. Answer: B (Knife : Cut)

Tool : primary function it performs. Pen writes, knife cuts.

Q8. Answer: B (Wolf : Pack)

Animal : its specific collective noun. Flock of birds; pack of wolves.

Q9. Answer: B (Malevolent)

Benevolent = wishing good. Opposite is malevolent (wishing evil). The others are synonyms.

Q10. Answer: B (Verbose)

Concise = brief; opposite = verbose (wordy). A, C, D are near-synonyms of concise.

Q11. Answer: C (Opaque)

Transparent = light fully passes through. Translucent = partial. Opaque = blocks light — the true opposite.

Q12. Answer: C (Descend)

Ascend = go up; descend = go down.

Q13. Answer: B

The passage weighs opportunities against challenges and closes on the need for policy frameworks — exactly the dual framing in B.

Q14. Answer: C

The passage names “algorithmic bias, data privacy, and the displacement of human labour” verbatim as critics’ concerns.

Q15. Answer: B

“Perpetuate inequalities” means to cause them to continue. Prefix *per-* (through) + root *petu-* (continue) signal maintenance.

Q16. Answer: C

The author presents critic and proponent views without endorsement, then identifies the policy challenge — hallmark of balanced/analytical writing.

Q17. Answer: C

Restates the closing sentence almost directly. A, B, D each contradict the passage.

Q18. Answer: B (Thorough)

Meticulous = showing great attention to detail; thorough is the closest synonym.

Q19. Answer: C (Unclear)

Ambiguous = open to more than one interpretation, i.e., unclear in meaning.

Q20. Answer: B (Persistent)

Tenacious = holding firmly; persistent is the standard synonym.

2. Analytical Reasoning – Worked Answers

Scenario 1 setup (Q21–Q24)

Positions 1–6 (left to right). D is at position 3, so A is at position 4. C is at position 1 or 6. The block B-between-E-F means EBF or FBE occupies three consecutive seats. F not adjacent to C forces the block to start at position 1: **E(1), B(2), D(3), A(4), F(5), C(6)**.

Q21. Answer: A (E)

Position 1 = E.

Q22. Answer: C (E and B)

Positions 1–2 are E–B, adjacent inside the EBF block.

Q23. Answer: A (D)

After swap: E, B, D, A, C, F. Fourth from right = position 3 = D.

Q24. Answer: A (One)

A at position 4, C at position 6 — only F (position 5) sits between them.

Scenario 2 setup (Q25–Q28)

P = Friday; R = Tuesday. The CT block is consecutive and cannot span Mon–Tue (R is on Tuesday), so the block must be Wed–Thu. D must precede C, so D = Monday. Order: **D, R, C, T, P**.

Q25. Answer: B (Design)

D = Monday.

Q26. Answer: B (Wednesday)

C = Wednesday.

Q27. Answer: B (Thursday)

T = Thursday, immediately following C.

Q28. Answer: A (D, R, C, T, P)

Direct read-off from the established schedule.

Scenario 3 setup (Q29–Q32)

Q = Java, S = JavaScript. $R \in \{\text{Python, Go}\}$. P cannot be Java or C++, and JS/Java are taken, so $P \in \{\text{Python, Go}\}$. R and P together cover $\{\text{Python, Go}\}$; T is left with C++.

Q29. Answer: C (C++)

Only language remaining for T.

Q30. Answer: C (Either Python or Go)

Constraint allows both; nothing pins R's choice.

Q31. Answer: C (Python)

If R takes Go, P — restricted to {Python, Go} — must take Python.

Q32. Answer: D (S)

Direct from constraint.

Statements (Q33–Q40)**Q33. Answer: B (Does not follow)**

Classic undistributed-middle fallacy. “Some graduates are professors” may refer to non-engineer graduates; the conclusion is not entailed.

Q34. Answer: A

The syllogism is formally valid given the stated premises, regardless of real-world truth about penguins.

Q35. Answer: A (QSPHSBN)

Each letter shifts by +1: $P \rightarrow Q$, $R \rightarrow S$, $O \rightarrow P$, $G \rightarrow H$, $R \rightarrow S$, $A \rightarrow B$, $M \rightarrow N$.

Q36. Answer: A (Father)

P is father of Q, and Q is sister of R, so P is also father of R.

Q37. Answer: B (42)

Differences: 4, 6, 8, 10, **12**. So $30 + 12 = 42$. (Equivalently $n(n+1)$ for $n = 1, 2, 3, 4, 5, 6$.)

Q38. Answer: D (Son)

“My mother’s only son” = the speaker himself (standard NTS interpretation for this template). The son of the speaker is the speaker’s son.

Q39. Answer: A (GCTVJ)

Shift +2: $E \rightarrow G$, $A \rightarrow C$, $R \rightarrow T$, $T \rightarrow V$, $H \rightarrow J$.

Q40. Answer: A

Valid syllogism. If all hardworking $\subseteq$ students, and no student is lazy, then no hardworking person is lazy.

3. Quantitative Reasoning – Worked Answers

Q41. Answer: C (100)

Number = $60/0.30 = 200$. Then $50\% \times 200 = 100$.

Q42. Answer: A (Rs. 680)

$800 \times 0.85 = 680$.

Q43. Answer: C (19)

Middle of five consecutive integers = mean = 17. So 15, 16, 17, 18, 19; largest = 19.

Q44. Answer: C (60)

$240/4 = 60$ km/h.

Q45. Answer: C (50)

$30/3 \times 5 = 50$. (Or: girls = $30 \times 2/3 = 20$; total = 50.)

Q46. Answer: B (Rs. 1200)

$SI = \frac{Prt}{100} = \frac{5000 \times 8 \times 3}{100} = 1200$.

Q47. Answer: B (25%)

Profit = 300; profit% = $300/1200 \times 100 = 25\%$.

Q48. Answer: C (12)

$24 = 2^3 \cdot 3$, $36 = 2^2 \cdot 3^2$. HCF = $2^2 \cdot 3 = 12$.

Q49. Answer: B (1/3)

$4/12 = 1/3$ of the job.

Q50. Answer: C (36)

$6 = 2 \cdot 3$, $9 = 3^2$, $12 = 2^2 \cdot 3$. LCM = $2^2 \cdot 3^2 = 36$.

Q51. Answer: B (12.5%)

If money doubles, interest equals principal: $P = \frac{P \cdot r \cdot 8}{100}$, so $r = 100/8 = 12.5\%$.

Q52. Answer: B (16 days)

Total work = $8 \times 12 = 96$ man-days. Time for 6 men = $96/6 = 16$ days.

Q53. Answer: B (4% decrease)

Net factor = $1.20 \times 0.80 = 0.96 \Rightarrow 4\%$ decrease.

Q54. Answer: A (49)

$$x = 7, \text{ so } x^2 = 49.$$

Q55. Answer: C (5)

$$2x = 10, x = 5.$$

Q56. Answer: B (2 and 3)

$$\text{Factor: } (x - 2)(x - 3) = 0.$$

Q57. Answer: C (25)

$$a^2 + b^2 - 2ab = (a - b)^2 = (3 - (-2))^2 = 25.$$

Q58. Answer: B (180°)

Standard Euclidean fact.

Q59. Answer: C (40 cm^2)

$$A = \ell \times w = 40.$$

Q60. Answer: B (44 cm)

$$C = 2\pi r = 2 \times \frac{22}{7} \times 7 = 44.$$

4. Subject – Worked Answers

D1. Physics (Q61–Q70)

Q61. Answer: B (Ampere)

Volt = potential difference, Ohm = resistance, Watt = power.

Q62. Answer: B (9.8 m/s^2)

Standard textbook value.

Q63. Answer: D (Speed)

Speed has only magnitude; the other three are vectors.

Q64. Answer: B (3 A)

$I = V/R = 12/4 = 3 \text{ A}$.

Q65. Answer: C (50 J)

$W = F \cdot d = 10 \times 5 = 50 \text{ J}$.

Q66. Answer: B (Refraction)

Reflection = bounce; diffraction = bending around edges; dispersion = splitting by wavelength.

Q67. Answer: C (Watt)

Joule = energy, Newton = force, Pascal = pressure.

Q68. Answer: C (Velocity)

Velocity has direction; the others are scalars.

Q69. Answer: A (50 Hz)

Pakistan and most of the world use 50 Hz; US/Canada use 60 Hz. 220 V is voltage, not frequency.

Q70. Answer: B (16 J)

$KE = \frac{1}{2}mv^2 = \frac{1}{2}(2)(16) = 16 \text{ J}$.

D2. Computer (Q71–Q80)

Q71. Answer: B (Operating System)

The others are application software running on top of the OS.

Q72. Answer: B (1024 bytes)

Binary KB = 2^{10} = 1024 bytes (standard NTS-conventional answer).

Q73. Answer: B (Modem)

MOdulator-DEModulator. Routers/hubs/switches operate on packets/frames, not modulation.

Q74. Answer: C (Control Unit and ALU)

Together with registers, these form the von Neumann CPU core.

Q75. Answer: C (HTML)

HTML is a markup language — it describes structure, not computation.

Q76. Answer: B

The defining feature is a name bound to memory whose contents may change.

Q77. Answer: B (//)

is preprocessor in C/C++ (and a comment in Python); /* */ is multi-line; <!--> is HTML.

Q78. Answer: B (Database Management System)

Canonical expansion.

Q79. Answer: C

Primary keys must be unique and non-NULL; the defining property is row identification.

Q80. Answer: D (SELECT)

INSERT writes new rows, UPDATE modifies, DELETE removes; only SELECT reads.

D3. Mathematics (Q81–Q90)**Q81. Answer: B (1/2)**

Standard trig value.

Q82. Answer: C ($3x^2$)

Power rule: $\frac{d}{dx}x^n = nx^{n-1}$.

Q83. Answer: C (11)

$2(4) + 3 = 11$.

Q84. Answer: C (3)

$1000 = 10^3$.

Q85. Answer: A ($x^2 + C$)

$\int 2x \, dx = x^2 + C$.

Q86. Answer: C (5)

$|a + bi| = \sqrt{a^2 + b^2} = \sqrt{9 + 16} = 5$. The classic 3-4-5 triple.

Q87. Answer: B (3)

$$m = (8 - 2)/(3 - 1) = 6/2 = 3.$$

Q88. Answer: B (24)

All four letters distinct: $4! = 24$.

Q89. Answer: C (2×4)

Inner dimensions match ($3 = 3$), so the product is valid with outer dimensions 2×4 .

Q90. Answer: C (2)

$$\cos 0^\circ = 1 \text{ and } \sin 90^\circ = 1, \text{ sum} = 2.$$

Compiled for NCSA / Air University.